SIMATS Engineering • CSE • CSA0804 — Python Programming • 14-08-2026 • Category: Class Work (11.8.26) • Student Submissions Report

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Q1. Selection Sort

Score: 0.00 TC: 0/3 passed

CODE

```
a = [23, 78, 45, 8, 32, 56]

n = len(a)

for i in range(n):
    min_index = i

    for j in range(i + 1, n):
        if a[j] < a[min_index]:
            min_index = j

    a[i], a[min_index] = a[min_index], a[i]

print("Sorted list:", a)
```

OUTPUT

```
Sorted list: [8, 23, 32, 45, 56, 78]
```

Q2. Insertion Sort

Score: 0.00 TC: 0/3 passed

CODE

```
a = [23, 78, 45, 8, 32, 56]

for i in range(1, len(a)):
    key = a[i]
    j = i - 1

    while j >= 0 and a[j] > key:
        a[j + 1] = a[j]
        j = j - 1

    a[j + 1] = key

print("Sorted list:", a)
```

OUTPUT

```
Sorted list: [8, 23, 32, 45, 56, 78]
```

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Q3. Merge Sort

Score: 0.00 TC: 0/3 passed

CODE

```
def merge_sort(a):
    if len(a) ≤ 1:
        return a

    mid = len(a) // 2

    left = merge_sort(a[:mid])
    right = merge_sort(a[mid:])

    result = []
    i = 0
    j = 0

    while i < len(left) and j < len(right):
        if left[i] < right[j]:
            result.append(left[i])
            i += 1
        else:
            result.append(right[j])
            j += 1

    result += left[i:]
    result += right[j:]

    return result


a = [23, 78, 45, 8, 32, 56]

print("Sorted list:", merge_sort(a))
```

OUTPUT

```
Sorted list: [8, 23, 32, 45, 56, 78]
```

Q4. Diagonal Matrix

CODE

```
n = int(input("Enter size: "))

matrix = []

for i in range(n):
    row = list(map(int, input().split()))
    matrix.append(row)

print("Diagonal elements:")

for i in range(n):
    print(matrix[i][i], end=" ")
```

OUTPUT

```
(no output)
```

Q5. Row and Column Sums

— Not Submitted —

Q6. Maximum Subarray Sum

— Not Submitted —

Q7. Common Elements Among Three Lists

— Not Submitted —